CHAPTER FOUR

The Social Structure of Work Groups

Looking back over the past several decades, one can note progressive changes in the focus of industrial research. Much of the empirical research on workers in industrial organizations has been motivated by an interest in practical problems of productivity. Earlier studies were chiefly concerned with physiological and economic factors relating to production. Researchers probed the nature of fatigue, experimented with pay incentive schemes, and examined the efficiency of worker efforts by means of meticulous time-and-motion studies. Taylor and his followers in "scientific management" attempted to dissect manual tasks in the hope of improving output by training workers to eliminate all body movements that were not essential, and they searched in vain for a set of "basic" activities to which all work behavior could be reduced.[1] As it became evident that scientific management's conception of workers as rational machines could not adequately account for their behavior, because it ignored the influence of sentiments and values, attention turned increasingly toward research on the underlying psychological processes that influence production. This psychological focus is best exemplified in studies of worker morale. But the morale of workers was found to depend at least as much upon the interrelations of the worker with his colleagues as upon his individual needs and dispositions. Hence, the recent trend has been away from an exclusive concern with the individual and toward an interest in the social factors in the work situation, beginning with the "human relations" approach in industry and expanding to include systematic research on the social structure of work groups.

These shifts in the focus of industrial research are rooted in changes in the organization of industry and in managerial ideology.[2] The development from small- to large-scale industrial enterprises has

1 A critical discussion of the Taylor school appears in James G. March and Herbert A. Simon, *Organizations,* New York: Wiley, 1958, pp. 12–22.

2 See Reinhard Bendix, "*Bureaucratization in Industry,*" Arthur Kornhauser *et al.* (eds.), *Industrial Conflict,* New York: McGraw-Hill, 1954, pp. 164–175. For a more detailed discussion of changes in managerial ideology, with cross-cultural comparisons, see Reinhard Bendix, *Work and Authority in Industry,* New York: Wiley, 1956.

been accompanied by a corresponding shift in the ideological justification advanced for managerial authority. In the days of the small shop, it could be argued with some plausibility that hard work and superior ability led to success, and the ideological assumption that the entrepreneur had demonstrated superior ability served to legitimate his authority over employees. But as industrial concerns grew in size and complexity, advancing within them rather than establishing a factory of one's own became increasingly the major avenue of success. With the development of the modern large-scale enterprise, moreover, the abilities the worker has occasion to demonstrate have ceased to lead to managerial success, since the latter depends on entirely different skills—not technical ones but skills in coordination, that is, skills in dealing with relations between people. The ideology consequently no longer glorifies primarily the old Puritan values of hard work and thrift, or even the later ones of entrepreneurial drive, but the new ones of personality salesmanship—of the ability to handle human relations effectively. The admonition "build a better mousetrap . . ." has changed into advice on "how to win friends and [that is, in order to] influence people." The idealized concept of the industrial organization as a "pyramid of opportunity" where advancement in a bureaucratic career is open to all who have the required social skills has replaced in large measure the earlier idealized concept of an industrial environment where every technically able employee could open up his own business some day. The problems of industry have come to be viewed in terms of human relations rather than in individual terms because "from the standpoint of the individual these techniques became a means of career advancement; from the standpoint of management they seemed to facilitate the coordination of a growing and increasingly specialized staff."[3] And this trend is reflected in parallel changes in research focus.

While the human-relations approach to industrial research assumes a more sociological perspective than its predecessors, it still is not adequate for the systematic study of social structure. It does avoid the atomization of work groups into their individual members and thus overcomes a limitation of most morale surveys. But the human-relations approach does not avoid a second pitfall: it tends to atomize the human relations in work groups by treating them as if they were attributes of individual group members, and it consequently ignores the organized network of social relations that characterizes group structures. This distinction may be illustrated by considering the relations between workers on an assembly line. Assembly-line workers, like others, establish friendly relations with co-workers nearby. But since they are strung out along the line, each person's set of interpersonal

[3] Bendix, "Bureaucratization in Industry," *op. cit.*, p. 174.

relations is different: a given worker will have contacts with workers on each side of him and with workers across the line, while his neighbor's contacts will include some of the same men, but not all, and will extend to others not included in the first worker's range of contacts.[4] On such an assembly line, therefore, we typically find human relations among workers but no work groups, since individuals are not involved in a *common* network of social relations set off from others by distinct boundaries. There are no definite subgroups within the larger collectivity with which workers can identify, and there are no subgroup norms defining common standards of conduct. In contrast to the assembly line, which inhibits the formation of work groups in the above sense, most industrial situations promote their formation. The study of the distinctive significance of group structure requires going beyond the human-relations approach to consider the networks of human relations and the common values which unite group members.

Informal Organization

A Pioneering Study. No single research has exerted more influence on the direction taken by students of industrial organization than the study carried out by Roethlisberger and Dickson in the Hawthorne plant of the Western Electric Company in Chicago.[5] A set of early experiments conducted in this plant had focused on the effects of varying degrees of illumination on worker productivity.[6] The first of these involved three different departments. When illumination was regularly increased, productivity increased in two of the three departments, but the increases were not parallel to the changes in illumination. A second study divided workers in one department into an experimental and a control group, with illumination being increased in the former and held constant in the latter; production increased in both of the two groups. A third study decreased illumination in the experimental group; again productivity of workers increased until the illumination had been reduced to a point where operators could no longer see their work. The conclusion seemed inescapable that the improved productivity noted was not due to increased illumination. The experimenters pointed out that many relevant conditions—including social ones—had not been controlled. Apparently what was involved in these strik-

4 See Charles R. Walker and Robert H. Guest, *The Man on the Assembly Line,* Cambridge, Mass.: Harvard University Press, 1952, pp. 67–68; and Abraham Zaleznik, *Worker Satisfaction and Development,* Boston: Graduate School of Business Administration, Harvard University, 1956.

5 F. J. Roethlisberger and William J. Dickson, *Management and the Worker,* Cambridge, Mass.: Harvard University Press, 1939.

6 *Ibid.,* pp. 14–17.

ing changes was that groups of workers routinely engaged in monotonous tasks were singled out for attention by management and by the researchers. This attention gave them a feeling of importance and made their jobs more interesting, and their consequent greater work satisfaction led them inadvertently to work faster.

Roethlisberger and Dickson, however, did not go so far in their interpretation at this point. They suggested only that the failure of the illumination experiments provided a stimulus for research in human relations and proceeded to set up the Relay Assembly Test Room.[7] Six girls were placed in a separate room (after their regular rate of production had been measured for two weeks prior to the transfer) where rest periods and other conditions, such as length of working day, could be varied. The experiment continued for more than a year. Regardless of the particular experimental variation in rest pauses introduced throughout the several phases of this experiment, the general course of the girls' productivity was upward. Several hypotheses were advanced to explain these surprising findings. For example, one possible explanation considered was that productivity increases were due to a reduction in fatigue produced by rest pauses. However, the productivity records for individual workers did not reveal the patterns that previous research had identified with fatigue, and hence this explanation was rejected. A second hypothesis was that the results obtained were due to changes made in the wage incentive factor.[8] To test this hypothesis, a second group of relay assembly workers were left in their regular work room but put on the same group incentive rate as that in the Test Room. Production increased in this group but far less than in the first group, a result the experimenters interpreted as indicating that although changes in earnings were important they did not account for all of the changes in productivity. Other explanations of the increase in productivity were similarly examined and dismissed. The conclusion finally arrived at was that increased productivity was a function of improved human relations. The entire social situation had been altered in ways that fostered friendly relations among workers. In addition, the supervision of workers had been taken over by the researchers who, in the interests of maintaining worker cooperation in the experiments, were very informal and nondirective in their approach.

Such findings as these led Mayo to reexamine some of his conclusions in an earlier study of a mule-spinning department in a textile

[7] *Ibid.*, pp. 19–186.

[8] From the third experimental period on, the test-room operators were formed into a separate group for the purpose of computing piecework earnings. This change was important since each operator had more control over her earnings in a group of six than in the large department. See *ibid.*, pp. 34–36.

mill.[9] Mayo had been called to this mill in 1923 because of the extremely high turnover in the spinning department. He introduced rest periods, and when this change led to improved productivity and lower turnover, he concluded that the rest period relieved postural fatigue induced by work and interrupted the pessimistic revery of workers. Later, however, he decided that the results obtained could better be interpreted in social rather than physiological terms. The workers gained status as they discussed their problems with researchers and company officials. Besides, and this change is most important, the men had been transformed from "a horde of 'solitaries' into a social group" as they consulted together on the setting of rest periods and used this free time for socializing.[10]

As the Hawthorne studies continued, an increasing awareness of the significance of social relations for worker morale led the investigators to decide to observe the behavior of a group functioning under normal circumstances rather than attempt to manipulate work conditions experimentally. An observer was placed in a room with 14 workers selected from a larger department; since the workers were engaged in wiring banks of telephone equipment the study came to be known as the Bank Wiring Observation Room.[11] It soon became apparent that there were uniformities in the behavior of the group under observation that did not follow the formal organization's blueprint. Informal relations developed among the men and gave rise to organized patterns of conduct in the group—that is, there was an informal organization.

The observer painstakingly recorded the overt manifestations of the network of informal relations that developed among the workers. There were distinct patterns of interaction: some workers frequently helped out certain others; games were regularly played at lunchtime which included some workers but not others. Sentiments of liking and respect were expressed primarily toward some group members, while others were not respected and were disliked. These observable aspects of their informal relations divided the workers into two cliques and a few isolates who were not members of either clique. It was among members of the same clique that most friendship ties developed, most games were played, and even most lunchtime conversations took place. While there was some conflict between cliques, there were social bonds that united the entire group and made possible the enforcement of common norms.

[9] Elton Mayo, "Revery and Industrial Fatigue," *Personnel Journal*, 3 (1924), pp. 273–281.

[10] Elton Mayo, *The Social Problems of an Industrial Civilization*, Boston: Graduate School of Business Administration, Harvard University, 1945, pp. 66–67.

[11] Roethlisberger and Dickson, *op. cit.*, pp. 379–524.

What group norms were there to enforce? In the course of interaction a set of common rules of conduct emerged, which included the following prohibitions: Don't be a rate-buster by working too fast. Don't be a chiseler by working too slow. If you are a straw boss, act like a regular guy; don't try to get bossy. Don't be a squealer. Conformity to norms was rewarded by approval that bestowed a relatively high position in the informal status structure. Norm violations were punished by group members in a variety of ways. Minor violations might be met with "binging"—striking the offender on the upper arm—or with ridicule. Continued violation of important norms resulted in a loss of popularity, a reduction in social interaction, and ultimately in complete ostracism. One worker was isolated because he violated the most serious group norm: he "squealed" on his fellows to the foreman.

Group norms also defined "a fair day's work," and this productivity standard fell below the level that management deemed desirable. These norms are discusssed by the authors under the label "restriction of output"; but this emphasis seems one-sided, since the norms discouraged overly low as well as excessively high production.[12] It is true, however, that workers guided by their informal standard of a fair day's work made less money than they could have. The fastest workers simply stopped working earlier than others to conform to group norms. And workers would discourage a fellow worker from working faster than the unofficial norm even though his output would increase their pay under the group incentive system. Such behavior seems irrational. But what is irrational depends on one's objectives. The workers' conduct can be considered irrational only if maximizing immediate income is assumed to be their sole objective, and this was not the sole objective.

The common objectives of these workers are implicit in the functions of their output norms. First, these norms allowed workers to increase their control over the environment and lessen their dependence on management. The workers were afraid that increased productivity would result in a cut in their piece rates, and although Roethlisberger and Dickson insist that none of the workers "had ever experienced any of the things they claimed they were guarding against" because Western Electric did not engage in such practices,[13] the workers had no assurance of this safety. Piece rates had been cut in other

[12] Although Roethlisberger and Dickson use the phrase throughout their work, they suggest at one point that the term "restriction" is perhaps not a good one, and that the informal control of production might be more accurately described as "behavior which was not strictly in accordance with the logic of efficiency." *Ibid.*, p. 537.

[13] *Ibid.*, p. 532.

companies,[14] and management had signed no contract with workers guaranteeing wage rates, there being no union at this time in the company. The workers' endeavor to control output was hardly irrational in these circumstances; indeed, it would have been irrational for workers to put their faith in the good will of management. The rational course was for the workers to take collective action in order to maintain productivity below the level that might tempt management to cut rates.

Norms controlling worker output also served the function of increasing job security for workers. The central objective of the workers was not maximizing their current income but attempting to keep their jobs so that they would have *some* income. For this study was conducted during the depths of the great depression. Millions of persons in the country were unemployed, there had been many layoffs at Hawthorne, and, in fact, the Bank Wiring Room study itself had to be discontinued because so many workers were laid off.[15] Thus, quite independent of possible changes in piece rates, an increase in productivity would have forced management to lay off workers in a depression if it were to act rationally. It would seem that restricting productivity was the most rational course open to workers to prevent layoffs.

A third function of informal output controls was to strengthen group solidarity by preventing competitive conflicts among workers. In the depression, each worker had strong incentives to keep his job. Roethlisberger and Dickson state that all but one of the workers participating in the study were in "very poor financial condition and if they were unemployed could not escape public support for long."[16] These circumstances could easily have led to cut-throat competition among the workers, each attempting to demonstrate that he, rather than his fellows, should be retained on the job because of his superior performance. The consequent conflicts would have torn the group apart and made the work situation most unpleasant had not norms controlling output maintained group solidarity by discouraging such rivalry. Given the situation in which workers found themselves, therefore, it appears that norms regulating the output of each worker were highly functional in protecting the interests of the entire group—they were rational means for this end.

[14] See S. B. Matthewson, *Restriction of Output among Unorganized Workers,* New York: Viking, 1931.

[15] Roethlisberger and Dickson, *op. cit.,* p. 385. Landsberger in his generally very favorable appraisal of the Hawthorne studies states that one of the authors' most regrettable errors was their failure to "draw attention to the fact that the rapidly worsening depression in all probability had an important influence on the atmosphere in the Bank Wiring Observation Room." See Henry A. Landsberger, *Hawthorne Revisited,* Ithaca: Cornell University Press, 1958, p. 58.

[16] Roethlisberger and Dickson, *op. cit.,* p. 531.

Informal Organization and Performance. Although the depression gave workers in the Bank Wiring Room a special incentive for controlling output, such adverse economic conditions are not necessary for the development of group norms regulating production. For example, Babchuk and Goode report a situation where a sales group developed a quota system that equalized sales volume for each member although management had established a commission arrangement encouraging competition among salesmen.[17] This unofficial system lessened competition among salesmen, improved their performance in nonsales areas, raised morale, and increased total group sales. A study by Roy of a group of workers in a machine shop also deals with regulation of output.[18] Roy's role as participant observer gave him access to practices that might well have been concealed from an outsider. Thus, he found that although workers deliberately restricted their rate of production, they often engaged in output races with one another. Such competitive games were concealed from management and did not affect the level of productivity since they were compensated for by goldbricking. Apparently, workers enjoy proving their skills through excellent performance and even relish competition among themselves; but they do so only when the stake in competition is limited and does not place in jeopardy the worker's earning power or his job. While quota restrictions give the workers collectively some self-determination over the conditions of their employment, they do so at the social cost of preventing the workers' pride in their skills and enjoyment of limited competition from making a contribution to operating efficiency. These conclusions imply that official procedures instituted by management to assure the employment security of workers might well improve operating efficiency; for they would obviate the need for informal regulation of output by workers and, thereby, free the inclinations of workers to strive to excel in their work.

The informal organization may contribute to as well as impede effective operations in the formal organization. It has already been mentioned that worker norms controlling production usually discourage too slow as well as too rapid work. Another important mechanism linking the formal and informal organization is the informal reward system. Among persons identified with their craft or profession, there is a tendency to look up to colleagues who are particularly expert in occupational skills. The respect and the popularity that such workers frequently enjoy are rewards for outstanding performance,

17 Nicholas Babchuk and William J. Goode, "Work Incentives in a Self-determined Group," *American Sociological Review*, 16 (1951), pp. 679–687.

18 Donald Roy, "Quota Restriction and Goldbricking in a Machine Shop," *American Journal of Sociology*, 57 (1952), pp. 427–442.

and the promise of informal status serves as an incentive to workers for becoming more skilled in their work. An association between competence and informal status has been observed in various kinds of work groups. For example, competence, as judged by the supervisor, was positively associated with being respected by colleagues in both the City and the County welfare agencies. The same positive relationship between competence and informal status among colleagues was found in an earlier study of a law-enforcement agency.[19] On the other hand, Homans concludes from a secondary analysis of the Bank Wiring Room data and other sources that high informal status goes with modal productivity, not with outstanding performance.[20] Other studies also have found that modal producers enjoy the highest informal status in work groups whose norms restrict productivity, because in these conditions the overproducers as well as the underproducers are penalized by loss of respect or even rejection.[21] It appears that the relationship between informal status and performance is contingent on work group norms: only if the expert exercise of skills is a dominant value in the group does high status tend to be associated with superior performance and to serve as an incentive promoting it; if the dominant norm standardizes productivity, high status is associated with modal performance.[22]

The cohesion of work groups often furthers operations. For example, cohesion has been shown to raise worker satisfaction and to lower turnover and absenteeism.[23] Cohesion also provides social support for workers; thus, it can neutralize the disturbing effects of conflicts with clients, as was noted in Chapter III. But the direct effects of cohesion on performance appear to be more variable. The findings of some studies suggest that cohesion promotes productivity. For instance, research on such varied groups as clerks, railroad employees, and factory workers found that workers in high-producing groups had greater pride in the accomplishment of their work groups (which may be considered an indication of cohesion) than did workers in low-

19 Peter M. Blau, *The Dynamics of Bureaucracy,* Chicago: University of Chicago Press, 1955, pp. 114, 117–121.

20 George C. Homans, *The Human Group,* New York: Harcourt, Brace, 1950, p. 141.

21 See, for example, Abraham Zaleznik *et al., The Motivation, Productivity, and Satisfaction of Workers,* Boston: Graduate School of Business Administration, Harvard University, 1958, pp. 231–232.

22 And in groups organized in outright opposition to the formal organization, such as are found in prisons or concentration camps, high informal status probably accrues to those members who can most effectively resist organization pressures; that is, to the "low producers" from the standpoint of the formal organization.

23 See, for example, Elton Mayo and George F. Lombard, *Teamwork and Labor Turnover in the Aircraft Industry of Southern California,* Boston: Harvard Business School (Business Research Studies, No. 32), 1944.

producing groups.[24] The researchers admit that the causal direction of the two variables measured is not clear, since a positive evaluation of the group's accomplishment may be either a cause or an effect of high productivity. However, they opt for the former because the workers had no objective information on their productivity as a group.

But other studies do not confirm this conclusion. For example, an experiment found that cohesion had no direct effect on productivity,[25] and so did a field study by Seashore of 228 work groups in a factory.[26] Both report, however, some indirect relationships between cohesiveness and productivity. Thus, Seashore did find, as he had predicted, that members of high-cohesion groups showed less variation in productivity than did members in low-cohesion groups, a datum which indicates that cohesion is associated with conformity to group standards. He also found that there was more variability in productivity *between groups* with high cohesion than between less cohesive ones. Finally, if the members of a highly cohesive group felt secure in their relations to the company, productivity tended to be high, but if they did not, it tended to be low. These findings suggest that cohesiveness increases the controlling power of the group over its members but that the direction in which this control is exercised—whether toward higher or lower productivity—is determined by other factors, such as the group's orientation to the organization.[27]

Informal Status. Informal status is not a unitary concept. Several dimensions of status have been identified, including power, prestige, and popularity. The various aspects of status are often but not always correlated.[28] One basic analytical distinction can be made at this point

[24] See Daniel Katz and Robert L. Kahn, "Some Recent Findings in Human Relations Research in Industry," Guy E. Swanson *et al.* (eds.), *Readings in Social Psychology,* New York: Holt, 1952, pp. 661–662.

[25] Stanley Schachter *et al.*, "An Experimental Study of Cohesiveness and Productivity," *Human Relations,* 4 (1951), pp. 229–238.

[26] Stanley E. Seashore, *Group Cohesiveness in the Industrial Work Group,* Ann Arbor: Institute for Social Research, University of Michigan, 1954, pp. 63–80.

[27] The work of Festinger and his colleagues tends to support these conclusions. In a study of housing groups, for example, they found that in the more cohesive groups there were fewer individuals who deviated from the group norm in their attitudes and behavior. Leon Festinger *et al., Social Pressures in Informal Groups,* New York: Harper, 1950. The study by Schachter and his associates (*loc. cit.,* see footnote 25), however, does not completely confirm these conclusions: high cohesion in the experimental groups studied was not consistently associated with more successful control of member conduct.

[28] In fact, Lenski has suggested that one important dimension of status is precisely the amount of agreement between the various status positions held by an individual. Gerhard E. Lenski, "Status Crystallization: A Non-vertical Dimension of Social Status," *American Sociological Review,* 19 (1954), pp. 405–413. See also Leonard Broom, "Social Differentiation and Stratification," Robert K. Merton *et al.* (eds.), *Sociology Today,* New York: Basic Books, 1959, pp. 429–441.

between two dimensions of social status in work groups: informal rank and social integration. An individual's informal rank is defined by the respect he and his opinions command among his fellows, and their consequent tendency to defer to him in social interaction. Rank is a hierarchical concept and refers to the relative standing of the members of the group. Integration is defined by social acceptance in a group, which is reflected in the degree to which an individual is drawn into interaction by his fellows. Although the integration of various members in a group often differs, it does not necessarily differ since all the members of a group may be highly integrated, or none may be. Thus, while rank is by definition a scarce commodity, integration is not and may be enjoyed by all group members, as is the case in highly cohesive groups.

We have already noted that the relationship between informal status and performance depends on the values and norms that prevail in a work group, and in a later section we shall analyze in greater detail the connection between informal status and group pressure. But let us now examine the implications of informal status for a person's orientation to others in the work situation—to his clients and to his colleagues. Caseworkers from the City Agency (which is described in the Appendix) were classified as being either primarily concerned with checking client eligibility for assistance in accordance with agency procedures (procedure oriented) or as chiefly interested in providing casework service to clients (service oriented). This distinction is based on each worker's description of how he had spent a day in the field visiting clients.[29] Social integration among colleagues, whether it was measured by popularity (number of friendship choices received) or by informal acceptance (being called by one's first name by other workers), was found to influence worker orientation, but only among the newer workers (see Table 6). Among workers with less than

TABLE 6. SENIORITY, INFORMAL ACCEPTANCE, AND ORIENTATION TO CLIENTS

Orientation to Clients	*Seniority*			
	More than 3 Years		*Three Years or Less*	
	Informal Acceptance		*Informal Acceptance*	
	High	*Low*	*High*	*Low*
Service oriented	50%	50%	54%	22%
N (= 100%)	10	8	24	18

[29] See Peter M. Blau, "Orientation towards Clients in a Public Welfare Agency," *Administrative Science Quarterly,* 5 (1960), pp. 341–361.

three years' experience, more than half of the highly integrated but less than a quarter of the unintegrated were oriented toward service rather than toward eligibility procedures, but among oldtimers (over three years' seniority), integration among peers did not affect orientation to clients. These findings can be interpreted by assuming that one factor that discourages casework service, although not the only one, is lack of familiarity with agency procedures, because it engenders anxieties that often find expression in rigid conformity with procedures. Several years of experience free workers from being preoccupied with procedures and thus make it easier for them to go beyond checking eligibility and provide some casework service. Half of the experienced workers did not provide much casework service, to be sure, but presumably the reason was that they were not interested in doing so rather than that anxious concern with procedures impeded their ability to do so. Social support from colleagues apparently relieves the anxieties typically associated with lack of experience, with the result that inexperienced workers who were integrated among peers were just as likely to furnish casework services as experienced workers. Only those workers whose anxieties were minimized neither by experience nor by social support were less inclined to provide such services.[30]

Now let us consider how informal status may affect relationships with colleagues. Caudill has shown that both formal and informal rank were related to participation in staff conferences in a private mental hospital.[31] With regard to formal position, senior doctors tended to participate most in the discussions concerning patients, residents were next highest in participation, and nurses and auxiliary workers, such as occupational therapists, participated least. Informal status rankings had an analogous effect on participation: the amount of participation of the various residents was directly associated with their competence as evaluated by their seniors and presumably with the respect they enjoyed among their colleagues.[32] Moreover, the outcome of a discussion varied according to the status of the individual raising the topic:

[30] Data from County Agency, using a different index of worker orientation to clients, essentially confirm these findings.

[31] William A. Caudill, *The Psychiatric Hospital as a Small Society*, Cambridge, Mass.: Harvard University Press, 1958, pp. 243–252, 295–296.

[32] Caudill's procedure was replicated in our study of County Agency. The weekly meetings of the division head and her supervisors were observed over a three-month period in two divisions. Caudill's findings on the relation between formal status and participation were supported; his findings on the relation between informal status (measured by the division heads' competence ratings of supervisors) and participation were supported in one division but not in the other. Besides, an earlier study by Blau found that the frequency of participation in staff conferences in a law-enforcement agency was directly associated with the social integration of the staff members. See Blau, *The Dynamics of Bureaucracy*, *op. cit.*, pp. 126–127.

on topics introduced by a senior doctor only 8 per cent of the discussions were inconclusive, whereas 14 per cent of the topics raised by residents and 21 per cent of the topics raised by nurses resulted in inconclusive discussions. Caudill notes that this situation evoked a sense of frustration among lower-status personnel.

How does a worker's standing among his colleagues influence his reference-group orientation? Data from City Agency again indicate that seniority in the agency was a crucial factor intervening between status and reference group. Workers were considered to define colleagues as their major reference group at work if they said that it was more important to them to be highly thought of by fellow workers than by their clients or by their supervisor. The pattern exhibited by the data in Table 7 was similar whether popularity, respect, or being

TABLE 7. SENIORITY, POPULARITY, AND REFERENCE GROUP

Reference Group	*Seniority*			
	More than 1 Year		*One Year or Less*	
	Popularity		*Popularity*	
	High	*Low*	*High*	*Low*
Colleagues	38%	15%	0%	41%
N (= 100%)	26	13	4	17

often consulted by colleagues was used to measure informal status.[33] Among newcomers (workers with less than one year of service) *low* status was associated with an orientation to peers, but among workers in the agency longer than one year *high* status was associated with a peer-group orientation.[34] It appears that newcomers who have already gained some popularity can turn their attention from their colleague group to concentrate on other problem areas associated with their jobs—their relations with clients and superiors. Newcomers who have not yet become popular continue to direct much of their energies toward achieving social status among peers. Older workers, on the other hand, have had time to adapt in some way to the social situation; if they have failed to win friends among peers they do not continue to look for social support to their colleagues but rather turn to clients or superiors. If old-timers have achieved popularity, however, the colleague group tends to

33 Whereas respect is clearly a measure of informal rank, and informal acceptance is a measure of integration, popularity refers in part to both of these dimensions.

34 Workers with one to three years' seniority were here classified together with those having over three years of experience because the response patterns of the two groups were similar.

be an important source of satisfactions for them and, hence, the reference group to which they often are oriented. In other words, it seems that unpopular workers lost interest in being highly thought of by colleagues over time (from 41 to 15 per cent), whereas popular workers became increasingly oriented to colleagues as their reference group with the passage of time (from zero to 38 per cent).[35]

Status relations are not confined to individual workers; groups also differ in status. The status of work groups is generally associated with their position in the organizational hierarchy. The standing of the groups to which a person belongs is one aspect of his own status, particularly in his outgroup relations, as is illustrated by the high informal ranks of the engineer who is part of management and of the secretary to the top executive. Changes in the wage structure may alter the relative standing of work groups; they therefore become a source of intergroup conflict. Whyte reports a controversy arising between work groups when the introduction of an incentive system in one department improved the take-home pay of a group of young machine operators. The older workers immediately demanded improvements in their incentive systems, transfers to better jobs, and other changes in an attempt to reinstate the former differential in their favor.[36] Both unions and management are continually faced with problems caused by changes that upset the established status order of work groups. Much of the resistance to change encountered in organizations is due to the disturbance which the proposed innovation would produce in the status structure.

Effects of Group Structure

Group Climate. The group climate or subculture is defined by the values and norms that prevail among group members. It is often asserted that the prevailing group climate influences individual conduct. But how do we know that the observed conduct is really the result of social pressures, since the individual may simply act in terms of his own attitudes and values? How can we distinguish between the influence of individual attitudes on behavior and the social constraints effected by group values and norms? To answer this question we must distinguish

[35] Our interpretation receives additional support from the fact that the tendency of workers to direct friendship choices to other members of their own unit rather than to the outgroup reveals roughly the same pattern as that in Table 7. Since we lacked these reference-group data for workers in County Agency, it was not possible to replicate the analysis.

[36] William F. Whyte *et al., Money and Motivation,* New York: Harper, 1955, pp. 67–70. The particular study reported is based on the research of Sayles.

two effects of group climate. First, the group climate can change the attitudes of individual members. We can observe this effect by noting the changes in attitude that occur in an individual after he joins a group, or we can infer such changes by comparing the attitudes of newcomers and of oldtimers as we did in the previous section. Second, and this effect is what concerns us now, the prevailing attitudes in the group can influence a group member's conduct *regardless* of his own attitudes. To isolate this type of structural effect, it is necessary to separate the external influence of group pressure from the internal influence of the individual's own orientation.[37] (The latter may have been brought by the individual to the group when he joined it, or it may be the result of earlier group pressure which he has internalized.)

The method used to separate structural effects from those of personality attributes can be demonstrated with data from City Welfare Agency. Twelve work groups were divided according to their prevailing group climate: groups where the majority of members favored a raise in the assistance allowance without qualification were considered to exhibit a proclient climate; work groups in which the majority of the members were opposed to raising the allowance or who tempered their endorsement by various qualifications were considered to have an anticlient atmosphere. (The alternative of lowering the assistance allowance was not favored by any caseworker.) *Within* these groups individuals were divided according to their attitudes toward raising assistance, using the same measure. In this manner it is possible to ascertain what effects of a proclient atmosphere on the individual's acting and thinking are independent of his own orientation, as the data presented in Table 8 illustrate.[38]

The first row of percentages in Table 8 indicates the significance of proclient values for service to clients. Proclient individuals were somewhat more apt than anticlient individuals to extend casework services to clients rather than merely to check their eligibility for assistance (compare individuals with different attitudes *within* groups). But regardless of the individual's orientation, he was more apt to render services to clients if he was in a group where proclient values prevailed than if he was in an anticlient group climate (note the difference between the first and third column, and that between the second and fourth column). The combined effect of group and individual values

37 See Peter M. Blau, "Structural Effects," *American Sociological Review,* 25 (1960), pp. 178–193.

38 It should be noted that this procedure does not control for the effects on the criterion variables of characteristics of individuals other than the specific counterpart of the structural variable. Neither does it control for the effects of other aspects of the group climate. Finally, Jack Sawyer has called our attention to the fact that failure to control the individual counterpart of the structural variable more effectively than by dichotomozing it may produce spurious structural effects.

TABLE 8. GROUP CLIMATE, INDIVIDUAL'S ORIENTATION, AND ATTITUDES

Attitudes toward Clients	*Group Climate*			
	Proclient		*Anticlient*	
	Individual's Orientation		*Individual's Orientation*	
	Proclient	*Anticlient*	*Proclient*	*Anticlient*
Service orientation	60%	44%	44%	27%
Willing to delegate responsibility	55%	78%	33%	50%
Worry little about cases	25%	56%	11%	32%
N (= 100%)	20	9	9	22

on service orientation was considerable: 60 per cent of the proclient individuals in proclient groups were service oriented, in contrast to only 27 per cent of the anticlient individuals in anticlient groups.

These data suggest that a worker's orientation influences not only his own performance but also his reaction to the performance of other workers in his group. If a proclient orientation prevails in a group, then the individual who merely checks client eligibility without providing casework services experiences disapproval, whereas the individual who provides services to clients earns the approval of his colleagues. The desire for social approval thus constrains workers in proclient groups to provide casework services to clients regardless of their own orientations to clients. In groups where the prevailing social climate is anticlient, social pressure works in the opposite direction, and the individual is socially rewarded for carefully checking eligibility but not for furnishing casework services. It is in this way that social processes—the distribution and direction of sanctions in interaction—exert an influence on a worker's approach that is independent of his own orientation to clients.

In respect to service orientation, the effect of a proclient group climate was in the same direction as that of proclient individual attitudes, and the two reinforced one another. But this reinforcement does not necessarily occur. The structural effect of the common value orientation in a group may be the reverse of the influence of the individual's internalized value orientation. This situation is presented by the data in the two lower rows of Table 8. The measure of willingness to delegate responsibility to clients is whether or not workers said that clothing should be made a regular part of the client's budget. At the time of

the study, recipients of public assistance at City Agency received a regular allowance for food and rent but not for clothing, which was supplied as needed at the discretion of the caseworker and his supervisor. In many other welfare organizations, including County Agency, a clothing allowance is included in the regular assistance budget, and City Agency workers were asked whether they would favor a change from their system to this system. Such a change would have saved them some tedious paper work, but it would also have deprived them of the power to furnish extra benefits to their clients at their own discretion, or to withhold them. A preference for the existing procedure implies that a worker is not willing to give up some of his own power of discretion in order to increase that of his clients. As Table 8 indicates, proclient individuals were less willing to delegate this responsibility to clients than were anticlient individuals. But the prevailing orientation in a group had the opposite effect: proclient group values increased the willingness of workers to delegate responsibility, both among proclient and among anticlient individuals. The last row in Table 8 reveals the same pattern. Workers were asked whether or not they tended to worry about their cases after working hours. Proclient individuals reported that they more often worried than did anticlient individuals; but membership in proclient groups operated to reduce worrying among both types of individuals.

Why is the influence of group values here the inverse of that of individual orientations? The following interpretation might be suggested: Proclient attitudes increase an individual's concern with his client's welfare, as indicated by the proclient worker's greater tendency to worry about cases. These same attitudes make him less willing to delegate clothing responsibility to clients, because he receives gratification from doing favors for clients, and because he can use his power to give clients more money than they otherwise could get by interpreting clothing needs liberally.[39] Furthermore, the extra burden of paper work is less objectionable for proclient individuals (who obtain satisfactions from helping clients that make the extra work involved worthwhile) than for anticlient workers (who do not experience such compensating rewards).

We may turn now to the structural effects. The prevalence of proclient values in a group probably gives rise to norms protecting the interests of clients. These norms tend to favor delegating responsibility to clients, partly to guard their interests against the actions of any workers

39 This interpretation implies that workers with proclient attitudes actually issued more clothing than others. This information is available only for a subsample of 17 workers. There is a difference in the expected direction, but it is too slight to be significant for so few cases; seven of the nine proclient workers and four of the eight others issued clothing frequently.

with anticlient attitudes, and partly to demonstrate confidence in and respect for clients. Norms that discourage too much worrying about clients are also more apt to develop in proclient groups, not only because emotional involvement is not conducive to optimum service to clients, but also because these peer groups must protect themselves against the inclination of their proclient members to worry excessively. Worried colleagues can make life unpleasant for those around them as well as themselves through their preoccupation with the misfortunes of clients and their tension. These problems confronted by proclient groups often give rise to agreement that the long-run interests of clients are best served by delegating responsibilities to them rather than keeping them dependent and by lack of involvement rather than feelings of great concern. In anticlient groups, on the other hand, there is no basis for such norms to emerge.[40]

Resistance against Group Pressure. The relationship between informal status and conformity has been widely discussed. Thus, Homans concludes that there is a direct relationship between these factors, because popularity and prestige are rewards for conformity to group norms and, conversely, low status is the penalty paid for deviancy.[41] Here the question under consideration is how conformity influences informal status. But we may also ask the opposite question: how does informal status influence conformity or resistance to group pressure. Homans suggests that, as the phrase *noblesse oblige* implies, high status constrains a person to conform meticulously to social norms. But other investigators have found that informal status—specifically, high integration or full acceptance in a group—promotes, at least under certain conditions, resistance to group pressures for conformity.[42] The latter findings suggest that the person not yet fully integrated has stronger incentives to comply with group standards in order to prove himself worthy of acceptance than the one who has already won the confidence of his peers.

[40] Data from County Agency permitted the replication of these tests. Service orientation could be replicated only roughly. County data revealed a structural effect in the same direction as that shown by City data but no individual effect was discernible. Data permitted the replication of the last two rows of Table 8 more precisely. City findings on willingness of workers to delegate responsibility to clients were exactly confirmed by County data. But County data did not confirm the findings on worrying, there being no difference between work groups in respect to worrying.

[41] Homans, *op. cit.*, pp. 140–144.

[42] See especially Harold H. Kelley and M. M. Shapiro, "An Experiment on Conformity to Group Norms Where Conformity is Detrimental to Group Achievement," *American Sociological Review*, 19 (1954), pp. 667–677; and J. E. Dittes and Harold H. Kelley, "Effects of Different Conditions of Acceptance upon Conformity to Group Norms," *Journal of Abnormal and Social Psychology*, 53 (1956), pp. 100–107.

Data from City Agency make it possible to test these alternative hypotheses—that informal acceptance among peers promotes conformity, or that it promotes resistance to group pressures.[43] For this purpose, both the twelve work groups and the 60 individuals in them were divided in terms of a given variable, just as was done in the previous section. Thus, groups in which the majority of workers reported that they often worried about cases were considered to have a "worrying" climate; the remainder of the groups were considered to have a "nonworrying" climate. Individuals within groups were then differentiated on the basis of this criterion into high worriers and low worriers. The measure of informal acceptance or integration used was whether or not an individual was called by his first name by some of the members of his work group. The data presented in Table 9 indicate that in groups where worrying prevailed, integrated workers worried slightly less than others, but in groups where few worried, integrated workers worried more than others.[44] Thus, in both kinds of groups, although their climates were the exact opposite of one another, integrated workers were more apt than others to deviate from the prevailing climate. In fact, it appears that integrated workers were not at all influenced by the group climate with regard to worrying: half of them worried in either situation—whether most of their co-workers did or most did not. In contrast, the group climate was clearly reflected in the orientations of the unintegrated workers, as indicated by the difference between 62 and zero per cent.[45] Essentially the same pattern of findings was observed if other measures of orientation were used, such as attitude toward raising the assistance allowance, and if popularity (received friendship choices) was substituted as the index of social integration.

The allegiance of workers to their supervisor revealed a similar pattern. Using as index of allegiance whether or not workers chose their own supervisor when asked which of the agency supervisors they would most like to work under, groups and workers were classified into loyal and nonloyal ones. In those groups where the majority of workers

43 See Peter M. Blau, "Patterns of Deviation in Work Groups," *Sociometry,* 23 (1960), pp. 245–261.

44 The form of Table 9 differs from that of Table 8 because now the individual's orientation (worrying) is assumed to be the dependent variable; this form simply makes it easier to read the table.

45 Data from County Agency did not permit an exact replication of this analysis of social acceptance and worrying, because the relatively low amount of worrying about cases reported made it difficult meaningfully to differentiate between work groups on this criterion. Work groups were divided using two or more workers who worried much to define a "worrying climate," and fewer than two to define a "nonworrying climate." When popularity was used as the index of social integration, the results presented in Table 9 were not replicated. However, when *perceived* integration ("Do you feel that you are a part of your division group?") was used as a measure, the results tended to confirm the findings in City Agency.

TABLE 9. GROUP CLIMATE, ACCEPTANCE, AND INDIVIDUAL'S ORIENTATION

Individual's Orientation	Group Climate: Worrying, Informal Acceptance: High	Group Climate: Worrying, Informal Acceptance: Low	Group Climate: Nonworrying, Informal Acceptance: High	Group Climate: Nonworrying, Informal Acceptance: Low
Worry much about cases	50%	62%	50%	0%
N (= 100%)	18	13	16	13

were loyal to their supervisors, highly accepted workers tended to be less loyal than unintegrated ones (73 per cent versus 100 per cent expressing loyalty). But in groups where few workers were loyal, the integrated workers tended to be slightly more loyal to their supervisor than the unintegrated (48 per cent versus 31 per cent expressing loyalty).[46] Thus, in both situations integrated workers deviated from the majority in disproportionate numbers. These findings as well as the previous ones tend to support the second hypothesis: social acceptance among peers seems not to promote conformity but to increase resistance against group pressure.[47]

Can these conclusions be reconciled with those of Homans? We believe that they can be inasmuch as Homans himself qualifies his thesis of *noblesse oblige* by introducing a factor which he terms "social security." He explains: "Up to a point, the surer a man is of his rank in a group, the less he has to worry about conforming to its norms."[48] We would add, "to *all* its norms," suggesting that it is necessary to discriminate between group norms on the basis of their importance. Norms that pertain to basic values of a group, such as output standards or the taboo on "squealing," are too significant to permit any member to violate them; hence, only outcasts are apt to do so. But some group norms are not this salient for the membership. The prevailing climates of opinion we have just analyzed are less important for group members and exert less severe restraints on them. So far as these group pressures,

[46] The numbers on which these four percentages are based are, in order, 11, 13, 23, 13. No data on the allegiance of workers to supervisors were collected in County Agency.

[47] Our data do not permit us to determine the direction of influence between these variables, but since it seems unlikly that deviation is rewarded by greater acceptance and popularity, the plausible assumption is that the direction of influence is the opposite.

[48] Homans, *op. cit.*, p. 144. For a more recent and fuller statement, see George C. Homans, *Social Behavior: Its Elementary Forms*, New York: Harcourt, Brace, 1961, pp. 339–358.

as distinct from the group's most salient values, are concerned, it appears that integrated members, who have already proved their acceptability, are permitted greater freedom to differ from the majority than unintegrated ones, who have yet to prove it.

Group Cohesion. Social cohesion refers to the strength of the network of social bonds that unite the members of a group. Its reference is thus clearly to group structure. While the feelings of integration or identification of individual members with the group are undoubtedly related to cohesion, they are not an appropriate measure of this structural property. A sociometric measure of group cohesion frequently utilized is the proportion of ingroup choices made by group members, but this index encompasses two distinct factors. A high proportion of ingroup choices means, first, that several, and perhaps most, group members are popular, that is, have established friendly ties with colleagues; and it shows, second, that all members find themselves in a group characterized by the prevalence of friendly bonds. While both of these factors clearly describe relations between people, only the second one is concerned with the structure of these relations in a group, and this attribute of group structure is the specific referent of the concept of social cohesion. Whether or not group cohesion in this narrow sense influences conduct independently of any influences the group member's personal social ties (his "human relations") exert can be answered by using the method of isolating structural effects.

The work groups in City Agency were divided on the basis of the proportion of friendship choices (best "friends" *in agency*) going to the ingroup, and individuals were divided by whether or not they received any choices from the ingroup, in order to examine the effects of group cohesion while holding constant the individual's ingroup popularity. To determine reactions to clients, the caseworkers were asked what behavior of their clients they found particularly trying. Some workers seemed to react largely in personal terms: they were upset when they felt insulted or exploited or when clients "made demands" on them. Others appeared to react in an impersonal manner: they objected to a client who engaged in behavior that they considered morally wrong or disadvantageous for the client himself or his family. Workers in this latter group might say, for example, that they found clients trying "when they do not take care of their children."

The data in Table 10 indicate that members of cohesive groups were less apt to react in personal terms to client behavior. Nearly two-thirds of them reported impersonal reactions, in contrast to only a quarter of the members of groups with low cohesion. Further, how popular an individual was in his work group had no effect on this aspect of his behavior. These results suggest that cohesive ties make the work group a source of strong social support for members. In the ab-

TABLE 10. GROUP COHESION, POPULARITY, AND REACTION TO CLIENTS

Reaction to Clients	*Group Cohesion*			
	High Individual's Popularity		*Low Individual's Popularity*	
	High	*Low*	*High*	*Low*
Impersonal	62%	66%	30%	20%
N (= 100%)	17	12	14	17

sence of such peer-group support, the caseworker's relations with his clients tend to become an important source of ego support for him, and his resulting dependence on clients leads him to react to them in personal terms. Group cohesion furnishes social support that makes relations with clients less significant for the caseworker and helps him to remain more impersonal toward them, while having personal ties with colleagues apparently does not furnish the social support needed for such independence from clients.[49]

Two other factors associated with cohesion may be briefly mentioned. The productivity of cohesive groups, as measured by total field visits per month, was slightly higher than that of other work groups. And the members of cohesive work groups were more likely than others to have respect for their supervisor (the index being that workers named him as one of the best supervisors in the organization). Perhaps cohesion creates a favorable disposition among workers toward the in-group and this attitude extends even to the supervisor. But perhaps the causal relation is the reverse: the fact that a supervisor has superior qualities, as indicated by the respect he commands, may be a condition that favors the development of group cohesion.[50]

THE LARGER ORGANIZATION AND WORK-GROUP STRUCTURE

Conditions of Group Solidarity. At this point we turn from the consequences to the determinants of group structure. We shall examine first some aspects of group composition that have been found to influence

[49] No data on personal versus impersonal reactions to clients were available from the study of County Agency.

[50] Data on field visits from County Agency did not replicate the findings from City Agency on the relation between cohesiveness and productivity. Data on respect for the supervisor were not available from County Agency.

group solidarity and then analyze the influence of the formal organization on the structure of work groups.

Group solidarity is a broader concept than cohesion. It encompasses not only the uniting bonds of group membership but also the collective strength derived from this unity. Indicators of group strength are to be found in cooperative activities and collective actions of various kinds. Sayles has pointed out that the significance of informal social ties among work group members has been exaggerated by Mayo and his followers. At least of equal importance is the degree of solidarity manifested in collective action and the accomplishment of common economic goals.[51] In his study, Sayles used the work group's union activity as an indicator of solidarity. Let us examine some research that explores the determinants of work group cohesion and solidarity.

There are indications that cohesive bonds are more likely to develop in homogeneous than in heterogeneous groups, that is, in groups whose members share interests and have a common background. This statement may be misleading, however, since individuals bring to their work group many characteristics and clearly cannot be alike in all of them. Besides, some differences will be more significant than others, and the salience of particular characteristics will vary from group to group. Systematic data often do not support the homogeneity hypothesis. Thus, Seashore found that variability in age and in education among the members of work groups were not related to cohesion.[52] And our data from the welfare agencies revealed no relation between group cohesion and homogeneity in age, in sex, or in religion. If anything, the mixed religious units in City Agency were more cohesive than those in which all members (or all but one) were Protestant. In general, it appears that heterogeneity in respect to those factors that are not particularly relevant for the work situation does not impede cohesion. Some aspects of heterogeneity, however, were found to be inversely associated with group cohesion and solidarity. In both agencies, for example, racially homogeneous work groups were more cohesive than were racially mixed units. In City Agency, moreover, cooperation[53] was greater in units that contained few newcomers and oldtimers but mostly workers with from one to three years' seniority. Sayles, having

51 Leonard R. Sayles, *Behavior of Industrial Work Groups,* New York: Wiley, 1958, p. 113.

52 Seashore, *op. cit.,* p. 82.

53 Cooperation was measured in terms of the willingness of the members of a work group to assume duties for which colleagues who were absent in the field were responsible. Of all clients who came to the agency when their worker was in, a certain proportion was seen; of those who came when their own worker was not in, a certain proportion was seen by other workers; the difference between these two proportions in a work group furnishes the index of cooperation (small differences show much cooperation).

examined 300 work groups in 30 plants but presenting no quantitative data, reports that occupationally heterogeneous work groups, whose members had diverse economic (union) interests, were less prone than others to engage in concerted union action.[54]

Research has also indicated that prestigeful background characteristics of workers are often associated with cohesiveness and solidarity. Seashore found that factory workers in cohesive groups had higher average seniority and were more likely to perceive their own job as having high status than those in other groups.[55] Zaleznik and his colleagues also report that the more solidary subgroups in the factory they studied were composed of workers possessing more prestigeful characteristics, including higher age, seniority, and education, and American or northwest European origins.[56] However, Sayles concludes that the work groups which exhibited most union solidarity tended to be composed of workers of medium status. He explains: "The highest prestige groups in the plant are so favored in their various endowments that they are less likely to exhibit open pressure tactics; the least favored are incapacitated for such tactics. Thus overt concerted interest behavior tends to be more concentrated in the 'upper middle groups' . . ."[57] The solidarity of the middle-status industrial groups may be the result of the ambiguity of the position of the semiskilled worker. The skilled occupations are similar in different firms, and hence their wage rates are determined by the market. The same holds for the unskilled. The positions of semiskilled workers seem to be more closely identified with a specific firm; these workers are often trained on the job and acquire skills that are not readily transferable. The worth of semiskilled labor is consequently ambiguous, since it is not determined by the market; the ambiguity gives these workers more leeway to improve their economic position and thus greater incentives to engage in collective action for this purpose.

Finally, the position of the work group in the larger organization's division of labor influences its social solidarity. Sayles found that groups engaged in tasks considered essential for the operations of the entire plant were more likely to exhibit union solidarity. The importance of the function performed increases a group's power and the chances for successful union action, and in this way promotes higher solidarity. Sayles also notes that union solidarity was lower in groups that performed interdependent differentiated tasks, such as those on assembly lines.[58] Two factors appear to be relevant here. First, in these

54 Sayles, *op. cit.*, pp. 59–61.
55 Seashore, *op. cit.*, pp. 87, 91.
56 Zaleznik *et al.*, *op. cit.*, pp. 69, 144–158, 198–212. But in the welfare agencies, group cohesion was not associated with high seniority.
57 Sayles, *op. cit.*, p. 55.
58 *Ibid.*, pp. 61–64, 83–84, 92–93.